



⌚ Time: 5 minutes

⌚ Outcome: Identify Minimum Efficient Scale & Constant Returns to Scale.

⌚ Difficulty: Intermediate

## Minimum Efficient Scale & Constant Returns to Scale



### THE CORE IDEA

Minimum efficient scale is the smallest output at which long-run average cost reaches its minimum. Before MES, additional scale can still lower long-run average cost; at MES the minimum has been reached. MES summarizes how much output a firm must reach before it captures all available scale economies.



### HOW TO RECOGNIZE IT

- Find the lowest point of the long-run average-cost curve.
- Minimum efficient scale is the smallest output at which that minimum cost is attained.
- A flat minimum indicates a range of outputs with constant returns to scale.
- Do not confuse a flat average-cost segment with output remaining unchanged.

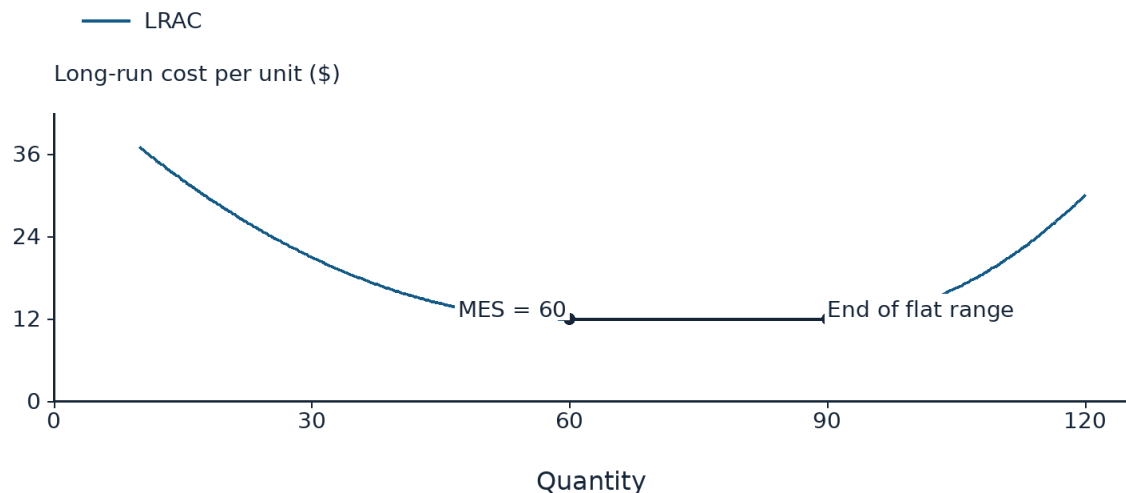


### WATCH OUT

Minimum efficient scale is the first output attaining minimum LRAC, not the midpoint or end of a flat minimum-cost interval.



### WORKED EXAMPLE: FINDING MINIMUM EFFICIENT SCALE



LRAC first reaches \$12 at 60 units and remains at \$12 through 90 units. Minimum efficient scale is therefore 60. The flat 60-to-90 interval illustrates constant returns to scale: output can expand proportionally without changing long-run cost per unit.



### CHECK YOURSELF

If LRAC first reaches its minimum at 5,000 units and remains flat until 8,000, MES is:?



**READY?**

**Return to the game and master this concept.**